



Do socioeconomic factors correlate with wind energy adoption in the United States?

Final Report

Handed in by:	Christoph Sommermann
Matriculation number:	22687554
Course:	Methods of Advanced Data Engineering

Research Questions:

1. Which US states have the highest number of wind turbines in the United States?
2. Which states have the most favorable socioeconomic conditions (e.g., lowest poverty rates, highest education levels)?
3. **Main question:** Do favorable socioeconomic conditions correlate with higher wind turbine adoption in the United States?

1. Introduction

Wind power is playing an increasingly significant role in the United States of America (USA), here are a few relevant facts. In 2021, the USA added 13.4 gigawatts (GW) of new wind power capacity and invested \$20 billion in wind turbines. That year, wind energy exceeded 20% of the state energy market in 11 states. In 2020, the U.S. generated 338 billion kilowatt-hours (kWh) of wind power, about 8.4% of total utility electricity production. Experts project wind power could become a relevant energy source nationwide by 2050, supported by its abundant availability (Parvin & Mayyas, 2023). Plausible factors influencing the adoption of wind power in U.S. states could include regulatory frameworks, geological aspects like wind speed, or the availability of usable land. However, these considerations are not the focus of this report. Instead, the goal is to examine whether wind power adoption at the state level correlates with socio-economic factors. To provide a clear answer, this report explores the following research questions using publicly available open data. *First, which US states are leading in the expansion of wind power plants. Second, which states have the best socioeconomic factors, such as low unemployment and high education. Third, is there significant correlation between these two factors.*

2. Used Data

This report utilizes two datasets, the **U.S. Wind Turbine Database**, and the **ACS 5-Year Socioeconomic Estimate Data**, both processed through the data pipeline described in detail in the data report. The U.S. Wind Turbine Database provides detailed information on wind turbines across the USA, spanning installations from 1982 to 2024. Key attributes include installation years (*p_year*), state locations (*t_state*), and specifications such as capacity in kilowatts (*t_cap*) and hub height in meters (*t_hh*). The dataset includes over 70,000 records and 29 columns. The ACS 5-Year Socioeconomic Estimate Data offers state-level indicators for the period 2016 to 2020, including median household income (*b19013est1*), poverty estimates (*b17019est2*), and employment statistics (*b20004Eest5*), representing the population aged 25 and over holding a bachelor's degree. It covers 52 states and territories with 144 columns detailing socioeconomic trends. Both datasets are in the public domain as U.S. Government Works and have been acknowledged in the references section of this report.

3. Analysis

The first research question, identifying the U.S. states with the most wind turbines, is addressed in Figure 1 and 2. Figure 1 highlights the top 10 states based on the cumulative number of wind turbines installed from 1982 to 2024, using state abbreviations for clarity. These states represent the leading regions in wind turbine adoption. The top 10 states with the highest number of wind turbines are **Texas** ('TX') with 18,651 windmills, **Iowa** ('IA') with 6,351 windmills, and **Oklahoma** ('OK') with 5,406 windmills, followed by **Kansas** ('KS'), **Illinois** ('IL'), **California** ('CA'), **Colorado** ('CO'), **Minnesota** ('MN'), **Oregon** ('OR'), and **North Dakota** ('ND'). Figure 1 was created using SQL queries on the processed windmill dataset, grouping by the columns *t_state* and counting *case_id* to calculate turbine totals per state. The data pipeline ensured a clean and standardized SQLite database for analysis. Figure 2 presents the analysis

geographically, highlighting the US states with the highest number of wind turbines in blue. The map was created using Python and Plotly.

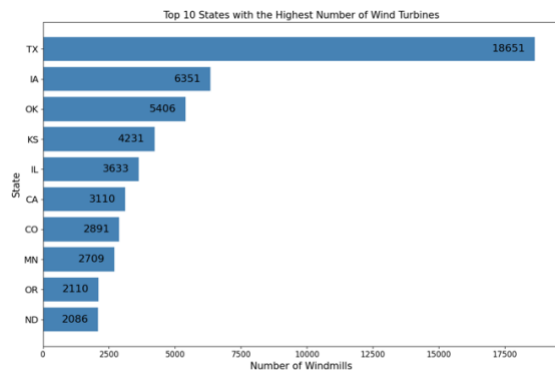


Figure 1: US-States With the Highest Number of Turbines



Figure 2: Map of Top 10 Wind Energy States

Figure 3 addresses the third research question by identifying the US states with the most favorable socioeconomic factors in the areas of poverty, income, and education. The figure comprises three bar charts, the first (green) highlights the states with the lowest number of households living in poverty, the second (blue) showcases the states with the highest median household incomes, and the third (purple) presents the top 10 states with the highest number of individuals holding a bachelor's degree.

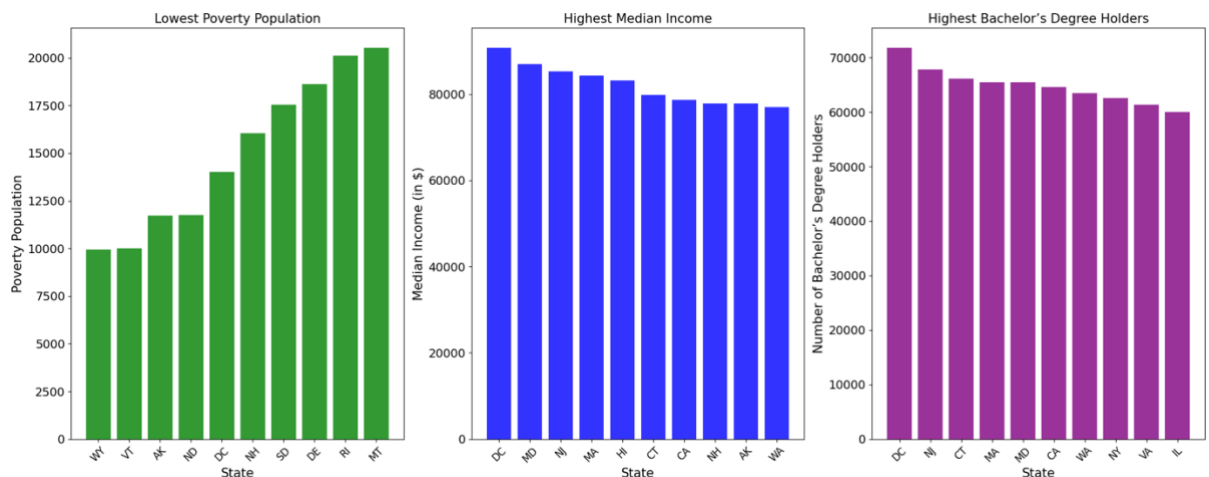


Figure 3: Top 10 States by Poverty, Income Ranking and College Education

The **green bar chart** shows that the top 10 states with the lowest poverty populations are Wyoming (WY), Vermont (VT), Alaska (AK), North Dakota (ND), Washington, D.C. (DC), New Hampshire (NH), South Dakota (SD), Delaware (DE), Rhode Island (RI), and Montana (MT). WY has the lowest poverty population with 9,939 individuals, while MT has the highest poverty population among the top 10 states, at 20,542.

The **blue bar chart** shows that the top 10 states with the highest median household incomes are DC, Maryland (MD), New Jersey (NJ), Massachusetts (MA), Hawaii (HI), Connecticut (CT), California (CA), New Hampshire (NH), Alaska (AK), and Washington (WA). DC has the highest median household income at \$90,842, while WA has the lowest median income among the top 10 states, at \$77,006.

The **purple bar chart** shows that the top 10 states with the highest number of individuals holding a bachelor's degree are DC, New Jersey (NJ), Connecticut (CT), Massachusetts (MA), Maryland (MD), CA, Washington (WA), New York (NY), Virginia (VA), and Illinois (IL). DC leads with 71,842 individuals, while IL has the lowest count among the top 10, with 60,045 individuals.

To answer the second research question clearly, the results from the three bar charts are compared and intersections are noted. The results of the comparison are shown in the Venn diagram in Figure 4. The diagram features three bubbles representing the bar charts from Figure 3, highlighting the states with the most favorable data in the areas of poverty, income, and education. The overlaps display the abbreviations of the states with the best overall socio-economic conditions across the factors considered.

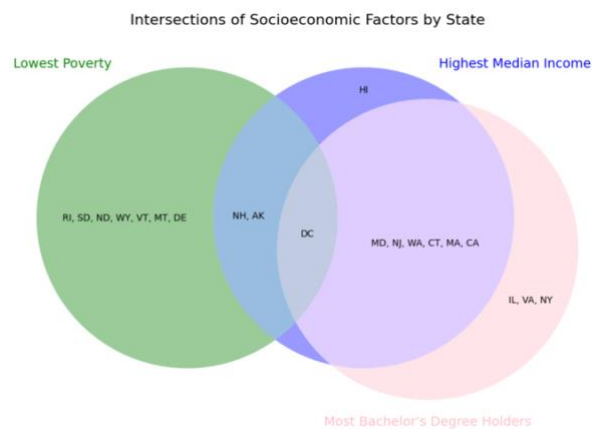


Figure 4: Intersections of Socioeconomic Factors by State

The diagram clearly answers the second research question, identifying DC, NH, AK, MD, NJ, CT, MA, and CA as the states with the best socio-economic conditions. To address the final and main research question — *examining the correlation between socio-economic factors and wind turbine installations* — the findings from the first two research questions must now be compared. Figure 5 presents a correlation matrix comparing the two lists of states. A value of 1 indicates an overlap between the lists, while a value of 0 signifies no overlap. This matrix provides a clear and efficient visualization of the relationship between socio-economic factors and wind power expansion. The matrix reveals that overlaps are almost nonexistent, with only California ('CA') showing a value of 1. As highlighted in Figure 1, CA ranks sixth in the number of wind power plants, and according to Figure 3, it also ranks sixth in both median income and the number of individuals with a bachelor's degree. The correlation matrix in Figure 5 indicates that there is no significant relationship between the expansion of wind power plants and socio-economic factors such as poverty, income, or education, thereby answering the third (main) research question. This indicates that favorable socio-economic factors do not drive the expansion of wind power plants, nor does a significant expansion of wind power plants lead to improved socio-economic conditions. Figures 3, 4, and 5 were created using Python with Matplotlib and Seaborn, visualizing top states for poverty, income, and education, their intersections, and the correlation matrix to analyze relationships with wind power expansion.

4. Conclusion

The present report analyzes three research questions using public datasets from the U.S. Wind Turbine Database and the ACS 5-Year Socioeconomic Estimate Data, both of which are publicly available and provided by U.S. government agencies. The first question focused on the U.S. states with the highest number of wind turbines, with Texas, Iowa and Oklahoma emerging as leaders in wind energy utilization. The second question looked at the states with the most favorable socioeconomic conditions, with Washington, D.C., New Hampshire, Alaska, and Maryland coming out on top. The third question examined whether socioeconomic factors correlate with the adoption of wind turbines. The analysis, including a correlation matrix, found no significant overlap between the states with the best socioeconomic factors and the states leading in wind turbine adoption, suggesting that there is no direct correlation.

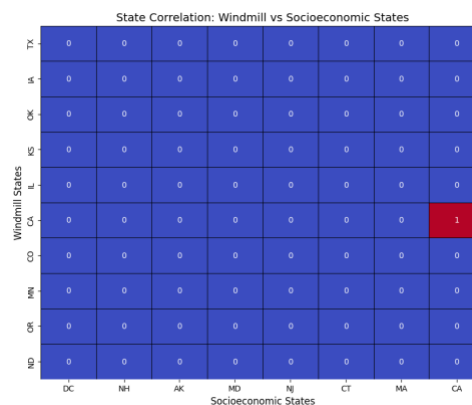


Figure 5: Correlation Matrix: Socio-Economic Factors vs. Wind Turbines Per State

Limitations

This final report contains different limitations. Firstly, the research results are based on only two data sources. This could have been extended to other data sources to identify possible differences between the data sources and possibly obtain even more accurate data. Secondly, the correctness of the data used could not be verified, this is because wind turbine data, for example, cannot be easily verified and therefore the information provided by the data issuer had to be trusted. This is a factor of uncertainty and a limitation of the work. A further limitation lies in the nature of the public data used; although the data was available online at the time the report was compiled, this can change at any time as the data source is online. This is also a source of uncertainty and could jeopardize the reproducibility of the analysis and research results if the data source is shut down.

References

Parvin, S., & Mayyas, M. (2023). Wind energy market in USA. *European Journal of Sustainable Development Research*, 7(1). <https://doi.org/10.29333/ejosdr/12538>

Data Licenses

U.S. Wind Turbine Database: The U.S. Wind Turbine Database is a U.S. Government Work, available for public access and use under the public domain. Proper acknowledgment of the data source has been ensured.

ACS 5-Year Socioeconomic Estimate Data: The ACS 5-Year Socioeconomic Estimate Data is also a U.S. Government Work, publicly accessible and in the public domain, with appropriate acknowledgment provided in this report.